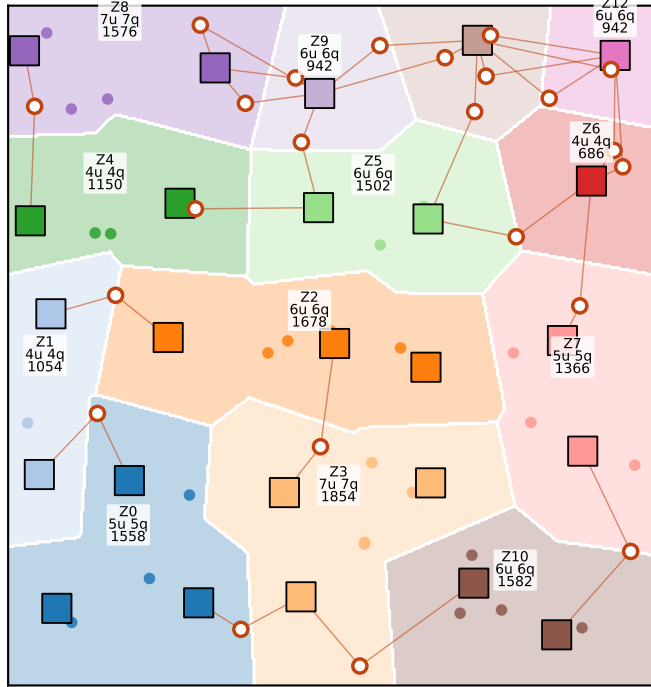


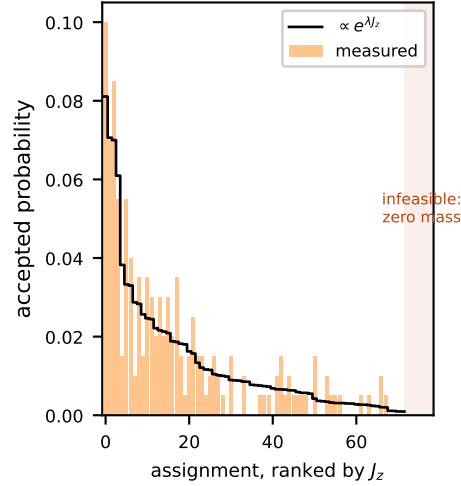
A region partitioned to the hardware budget, solved by zone-local sampling and classical boundary coordination

(a) partition under a 2000 two-qubit budget
25 APs, 50 UEs $\rightarrow$ 13 zones, $|\mathcal{B}|=23$ boundary UEs (46%)

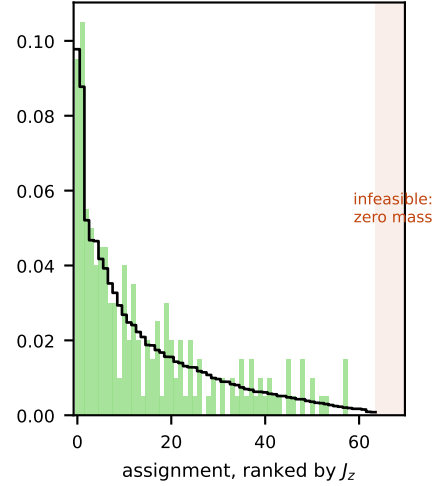


shading: zone territory (nearest AP)
 AP (colour = zone) interior UE
 boundary UE UE-to-candidate-AP link

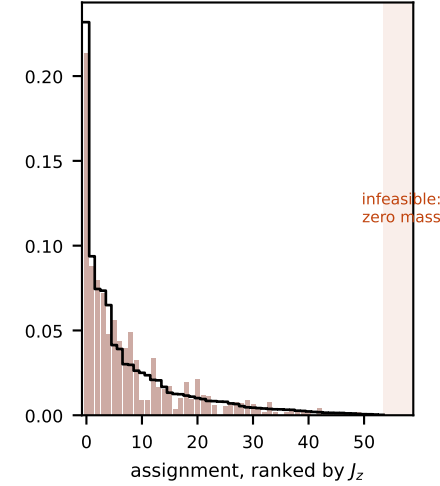
zone Z3: $N_z=7$, $|\mathcal{F}_z|=72$
 $\mu_z=0.096$, 2k shots, ESS 200/200
 executed at target β



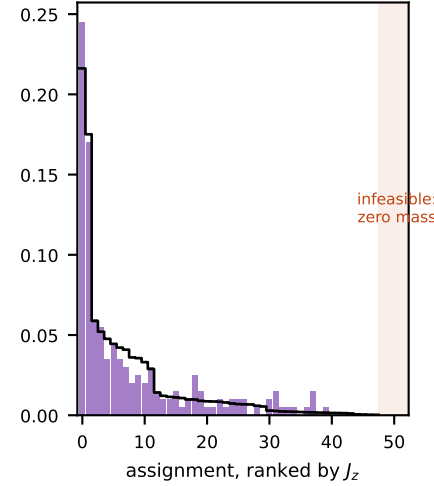
zone Z5: $N_z=6$, $|\mathcal{F}_z|=64$
 $\mu_z=0.160$, 1k shots, ESS 200/200
 executed at target β



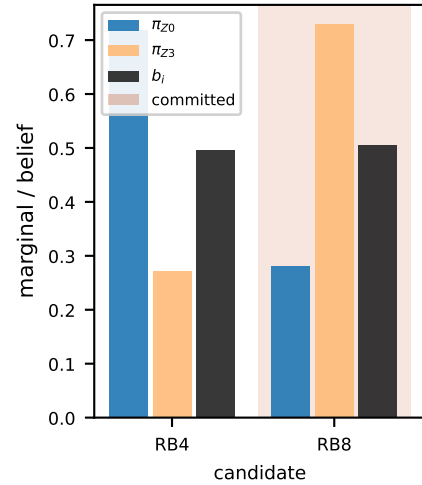
zone Z11: $N_z=7$, $|\mathcal{F}_z|=54$
 $\mu_z=0.020$, 10k shots, ESS 150/200
 executed at $\beta_z=0.85$, reweighted



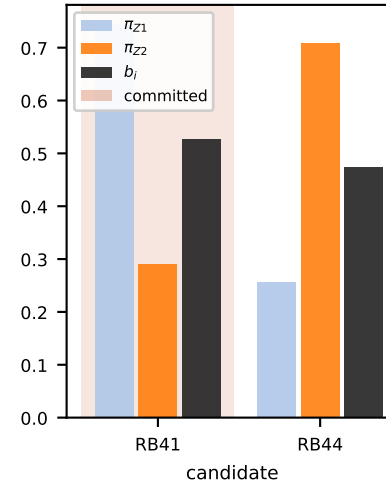
zone Z8: $N_z=7$, $|\mathcal{F}_z|=48$
 $\mu_z=0.072$, 3k shots, ESS 200/200
 executed at target β



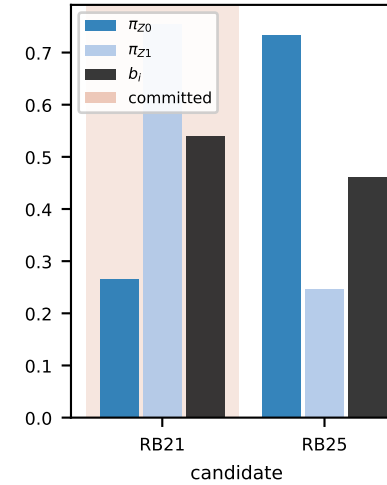
boundary UE 29: Z0 + Z3
 confidence 0.50



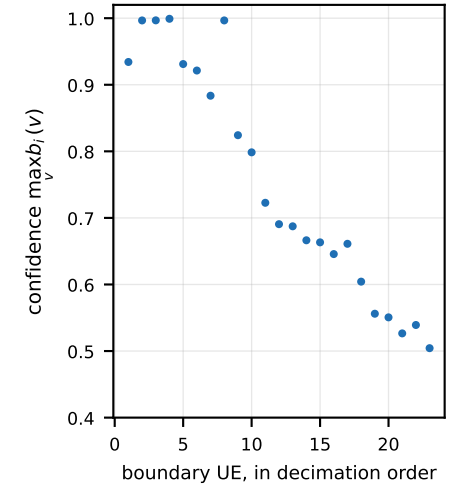
boundary UE 25: Z1 + Z2
 confidence 0.53



boundary UE 16: Z0 + Z1
 confidence 0.54



each boundary UE fixed once,
 least ambiguous first (5 exception re-samp)



$g=5$, seed=0 (median of 8 seeds by utility ratio); target beta=1.5, $K_z=200$, shot budget 10,000/zone (1 zones executed at a lower beta_z and were reweighted). Utility 297.2 of the centralized strict optimum 299.1 = 99.4%; the assignment is strictly feasible (True). Zone labels are exact: the accepted branch of the circuit matches $\exp(\lambda J_z)$ on $\mathcal{F}_z$ to statevector precision (max TVD $2.7e-15$ over the zones checked in haiq_certify.py), so the sampler used here is the same law the circuit produces.